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Understanding Preservice Teachers' Practices of Embedding Play-Based Learning and Gender-Responsive Pedagogies During Practicum

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ABSTRACT

This study aims to understand preservice teachers' practices of play-based learning (PBL) and gender-responsive pedagogies (GRP) during the practicum in their teacher education program. We used a qualitative case study design including interviews and observation with 17 preservice teachers who were taught by teacher educators attending professional development programs on PBL and GRP across four Palestinian universities. We conducted thematic analysis and then mapped the findings according to the COM-B model. Preservice teachers showed strong capability and motivation in using PBL, but had limited understanding of GRP, which reduced their capability to apply it despite the application of some equitable practices. Preservice teachers' opportunities to implement PBL were strongly shaped by their physical classroom conditions and by the guidance and modeling provided by their teacher educators. Whereas limited exposure to GRP left preservice teachers uncertain about its relevance and application. These findings highlight the importance of improving teacher educators' practices through professional development programs to improve teacher education programs. Further investigation is required for a better understanding of preservice teachers' practices of GRP.

Introduction

Play-based learning (PBL) is a student-centered approach to education which uses play as a tool for building children's knowledge and skills (UNICEF, 2018). PBL does not have a strict definition and continues to be an area for debate in research. PBL can include free play, guided play and playful direct instruction (Zosh et al., 2018). This broad definition of play-based learning gives teachers a wide number of choices when implementing this pedagogy (Pyle & Danniels, 2017). Evidence suggests that PBL can be valuable in higher education in terms of motivation and engagement (James 2019;

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Whitton, 2018). For example, PBL is an essential component in engineering education for undergraduate and master students at the University of Pennsylvania, where the imaginative and intrinsically motivating values of play are foregrounded within courses (Kothmann, 2019). Similarly, using a board game to teach brand elements (related to marketing) to undergraduate business students at an Australian university improved students' motivation, enjoyment and absorption, (i.e., "flow" (Khan & Pearce, 2015). PBL provides meaningful experiences which help undergraduate students improve their skills, knowledge, and aptitude to be engaged citizens and has proved beneficial to preservice teachers' enjoyment while studying and professional readiness for their future work (Jensen & Rørbæk, 2022; Sinfield et al., 2019). In order to facilitate PBL's vital role in the learning process, skills for using PBL in the classroom across a diversity of subjects should be included in both pre-service and in-service teacher education (Diaz-Varela & Wright, 2019).

In addition to using play to stimulate learning, PBL may also be used to support gender equity in the classroom through the complementary use of gender-responsive pedagogies (GRP). GRP has been described as "teaching and learning processes which pay attention to the specific learning needs of girls and boys" (UNESCO Bangkok, 2017, p. 4). The potential benefits of applying gender-responsive pedagogies for students include eliminating gender stereotyping, improving active participation for all students, and developing the skills required to compete in the workplace (Nabbuye, 2018). GRP supports transformative learning by helping the learners to become aware of society's gender assumptions and enabling educators to examine the learning materials that reflect real-life experiences and ensuring equal participation for all genders in group work (Skovgaard & Chapin, 2021). GRP promotes gender equality in educational contexts, where gender equality refers to the equal rights, opportunities, and responsibilities for individuals regardless of their gender identity, achieved by eliminating gender-based discrimination and promoting inclusive society (Alemayehu, 2020; Moore, 2024). According to Lamptey et al. (2015), promoting gender equality in teacher education involves a broad range of policies and practices including creating gender-responsive plans and policies, collaborating with institutional cultures and environments, working to change teacher attitudes, and developing gender-responsive instructional materials and pedagogy. It is necessary to evaluate and create a teacher education curriculum which responds to the aims of gender equality. Pedagogically driven gender inclusion should be integrated to support gender equality in higher education effectively (Kahamba et al., 2017; Kreitz-Sandberg, 2013).

In this regard, we conducted a four-year research project which aimed to embed PBL and GRP in teacher education programs in Palestine. The project developed out of an existing research collaboration between the two PIs (Aljanazrah & Murtagh) based at the University of Birzeit and University of Limerick which was supported initially by Research Ireland's New Foundations grant with the subsequent awarding of a four-year COALESCE (Collaborative Alliances for Societal Challenges) grant to complete the research described herein. Dr. Aljanazrah is an educator from Palestine with years of experience in developing teacher educator programs and Prof. Murtagh has experience working in education in Palestine as well as teacher education in Ireland focusing on questions of equity and social justice. From previous research collaborations and experience using PBL in diverse educational settings, both Dr. Aljanazrah and Prof. Murtagh believed that a greater uptake of PBL and GRP could make a

positive impact on the learning experience across all levels of the education system and were keen to see how this could be further embedded into teacher education. Two local researchers from the West Bank were hired to support the project: Dr. Nibal Khalil, was hired as a postdoctoral researcher who led the data collection and analysis, alongside Ghadeer Hamed who was hired as a research assistant. The research team relied on existing relationships with Deans of Education and teacher educators for approaching the four participating universities and recruitment of participants for interviews and observation. In the final nine months of the project, Dr. Aisling Walsh, an Irish based researcher, was hired to replace Dr. Khalil supported the team with the analysis of the GRP aspects of the program. The team was therefore composed of researchers with insider knowledge and expertise in both teacher education, as well as the context of Palestine, and two Irish researchers who were familiar with the context but could also provide an outside perspective.

The first phase examined perspectives on PBL and GRP with teacher educators from higher education institutions (Khalil et al., 2022, 2024). Based on these results, the second phase included the design of a professional development program to meet teacher educators' needs toward embedding PBL and GRP in their teaching practices. This professional development program was then implemented with teacher educators, and we found subsequently that they had developed the appropriate knowledge and skills to adopt PBL and GRP in their teaching practices within the physical and social environments of the classroom (Murtagh et al., 2024). This paper corresponds to the third phase of the project where we sought to understand preservice teachers' practices of embedding PBL and GRP in their teaching during their practicum modules. The preservice teachers who participated in the study had been taught by the teacher educators who attended the professional development program for PBL and GRP implemented during Phase 2. Results will offer new insights into enhancing PBL and GRP in teacher education programs and will demonstrate to relevant stakeholders, such as the Ministry for Education and Deans of Education at the participating universities, the importance of teacher educators' professional development programs in developing preservice teachers' practices and improving the quality of teaching. Additionally, results will provide new evidence of the importance of enhancing the practices of teacher educators by using innovative teaching methods in higher education.

Role of preservice teacher education

Preservice teachers acquire theoretical and practical knowledge from initial teacher education programs which influences the quality of teaching (Musset, 2010). Given the importance of using innovative pedagogies in teaching and learning at schools, embedding these pedagogies within teacher education encourages educators to inspire, engage, and empower learners for the changing world (Petrenko, 2024). Active learning enhances preservice teachers' abilities to incorporate dignity, energy, self-management, community, and awareness into their teaching practice (Olgun, 2009). Thus, preservice teachers should be well-trained during their participation in teacher education programs in using innovative teaching methods to ensure quality education for the next generation. Preservice teachers are required to improve the content knowledge, pedagogical knowledge, practical teaching experience, using various teaching resources, and engaging in specialized training to increase their confidence in teaching science and math for

students (Chen et al., 2022). Another factor that enhances preservice teachers' performance is improving teacher educators' knowledge and skills through qualification and professional development programs that allow them to reflect on the use of research and teaching with empirical evidence (Diery et al., 2020). Many of the teaching methods used in higher education are still traditional (Mandal, 2018) and when preservice teachers begin working professionally, they often apply those same methods (Postareff & Lindblom-Ylänne, 2008). There is a great need for professional learning opportunities for teacher educators, so that evidence-informed strategies, pedagogies and practices can be incorporated into teacher education programs. A growing body of research explores the relationship between teacher educators' practices and the impact of the learning experiences of preservice teachers. For example, Chan et al. (2023) revealed a positive relationship between teacher educators' instructional behaviors (i.e., instructional clarity and support for student autonomy and cooperative learning) and preservice teachers' intrinsic learning motivation and teaching self-efficacy. Moreover, Gravett and van der Merwe (2023) argue that preservice teachers benefit when teacher educators model a learner-centered lesson design approach in which the rationale for planning decisions is explained. Furthermore, Mikeska et al. (2023) investigated the value of teacher educators and their preservice teachers in using simulated teaching experiences. The findings stressed the importance of teacher educators using techniques to engage the preservice teachers in learning how to develop core practices of facilitating discussions. More research is needed to explore the influence of teacher educators' pedagogical approaches on pre-service teachers' practices, to further enhance the quality of teacher educator programs.

According to factors that affect preservice teachers' pedagogical practices, previous research from a Hong Kong-based initial teacher education program demonstrated a relationship between intrinsic motivation and preservice teachers professional learning, particularly the intellectual aspect of their professional competence, including subject matter, pedagogical and educational knowledge through their learning in teacher education and their interaction with others (Tang et al., 2020). According to self-determination theory, intrinsic motivation leads people to engage in a specific behavior because of inherent satisfaction, interest in and enjoyment of those things, and the potential for improving their skills (Ryan & Deci, 2000). Intrinsic motivation is reflected in the fact that early childhood preservice teachers believe that play is not only enjoyable for students but also for themselves as teachers (Altun, 2018). Preservice teachers' motivations to learn are influenced by their perceptions about the benefits of the course content for their future careers as teachers and for promoting self-regulated learning strategies (Lee & Turner, 2017). The factors which impact motivation such as empowerment, usefulness, success, interest, and caring, could be supported using active learning assignments which lead to high-quality teachers (Caruso, 2021).

When it comes to understanding PBL, not all preservice teachers define play in the same terms, instead, they create their unique definitions based on their experiences and shaped by their beliefs (Sherwood & Reifel, 2010). Two principal types of PBL have been identified: free play and guided play. Free play is child-led and driven by children's interests which contributes to their development across physical, social, emotional and cognitive domains. This highlights the importance of teachers' understanding of the links between free play and learning (Santer & Griffiths, 2007). Guided

play, on the other hand, blends the child-led nature of free play with a focus on specific learning outcomes under adult guidance (Weisberg et al., 2016). The teacher's role in guided play varies, with the degree of teacher engagement depending on the subject matter, learning environment, student needs, and materials (Pyle & Danniels, 2017). Preservice teachers have identified various roles for teachers when applying guided play including planner/organizer, observer/guider, play partner, and non-involver (Altun, 2018). Therefore, preservice teachers should be supported in embedding PBL activities into future teaching practices. This includes developing an understanding of the connections between free play and learning, as well as identifying the suitable teachers' role (e.g., planner, guide, partner, and non-involver) when implementing PBL activities within educational contexts during their teacher education. This study aims to explore preservice teachers' implementation of PBL with students and examine how they perceive of and enact their roles within PBL activities.

The United Nations Children's Fund (UNICEF) global Strategic Plan, 2022–2025, identifies gender-responsive education systems and equal access to education for all as a key action for achieving its goals on gender equality (UNICEF, 2021). Although teachers emphasize the importance of GRP in teaching and learning, the implementation of GRP in the classroom remains low (Abd Malik et al., 2023). Therefore, universities should develop strategies whereby academics, in general, and teacher educators, in particular, can implement GRP during their teaching practices individually and collectively (Kreitz-Sandberg, 2013). Several studies, however, have illustrated that teachers need to improve their competence in this area. For example, a study in Bhutan demonstrated that teachers' lack knowledge about gender issues or an understanding of GRP which hinders their ability to incorporate GRP into their lesson plans in a higher secondary school (Dorji, 2020). In Ethiopia, Alemayehu (2020) identified the challenges of embedding GRP at a governmental university showing that, although the university teachers had some awareness of GRP, they did not use GRP in their lesson planning due to the lack of training. Similarly, their students responded that most educators did not consider equal participation in the classroom, instead providing more opportunities to the dominant students in the class (male students in this case). Our study in Palestine revealed that teacher educators have insufficient knowledge of gender issues and skills to embed GRP in their teaching (Khalil et al., 2024). Therefore, greater awareness and sensitivity toward gender equality should be developed for students, teachers and preservice teachers to reduce the discrimination and inequality practices in the educational contexts (Şanlı & Karakuş, 2023).

According to Ananga (2021) and Schreurs and Rundgren (2023), education on gender equity should be included in teacher education programs. A further study from a state university in Ankara, Turkey found through pre/post-test analysis that the 33 preservice teachers (out of 100) that this course had a substantial impact on the attitudes of preservice teachers in supporting greater gender equity in the classroom (Erden, 2009). Another study in Ghana conducted by Ananga (2021) showed that teacher educators who received specific training on gender, increased their use of GRP for teaching core subjects (English Language, Mathematics and Science) from 2015 to 2018. Despite this improvement, there were some challenges in implementing GRP in their practices including limited resources, for example a lack textbooks and reading materials that supported GRP, class size and the lack of gender diversity in classes

(less than 10% male students). Not only do preservice teachers have to learn how to embed gender equality in their practice, but they must also learn how to teach it to the students (Schreurs & Rundgren, 2023). Ananga (2021) argues that it is essential to prepare future teachers to deal with the various gender stereotypes and biases which students bring from home through Ghana's initial teacher training program. The Forum for African Women Educationalists (FAWE) designed a GRP model as an effective intervention for enhancing the promotion of gender equity and equality in education in general, and girls' performance in particular (Jahangir & Mankani, 2016). This model was implemented in primary and secondary schools and teacher training colleges in six countries: Malawi, Tanzania, Uganda, Ethiopia, Gambia, and Zambia. The authors concluded that the best option for scaling up GRP on a national level was through preservice training.

Theoretical framework

The Capability, Opportunity and Motivation - Behavior (COM-B) model (Michie et al., 2014) is a theoretical framework which aims to understand behavior change. According to this model, capability refers to an individual's psychological and physical ability to engage in any given activity, including their knowledge as well as physical and mental skills. Opportunity refers to external factors that affect behavior including the physical and social environment. Lastly, motivation refers to the conscious and unconscious cognitive processes that direct and inspire behavior. These processes can be reflective and involve personal goals, beliefs, and making plans, and can also be automatic involving emotions and habits (Ibid). The COM-B model has been used successfully in educational research to understand key aspects of teacher behavior and design educational interventions. For example, the UK Department for Education used the COM-B model as a behavioral lens when designing school support services (Department for Education, 2022), while in Sweden it was used when examining staff perceptions of implementing a framework for inclusive behavioral support in schools (Fohlin et al., 2025). The model guided the thematic analysis of interviews with 5th grade school-teachers regarding the implementation of in-classroom physical activity breaks in Sri Lanka (Peiris et al., 2023). At 3rd level the COM-B model was used to identify the main determinants for the use of learning design methodologies through structured interviews with staff from UK universities (Toro-Troconis et al., 2021) and to examine AI adoption attitudes in the Southern African higher education sector (Patterson et al., 2025).

Palestinian context

The educational context in Palestine is marked by a protracted history of social and political conflict and upheaval which has impacted access to education, despite the prioritization of education in Palestine and the support from development agencies such as international donors and United Nations Relief and Works Agency for Palestine Refugees in the Near East (UNRWA) (United Nations Palestine, 2023). UNRWA is a United Nations (UN) agency, established in 1949, that provides essential services – including education – for Palestinian refugees in Gaza, West Bank, Jordan, Lebanon

and Syria. The three types of schools in Palestine include governmental schools (74% of the total number of schools), private schools (12%), and UNRWA schools (15%). While overall literacy rates in Palestine are among the highest in the world, at 97.7%, school completion rates, particularly for boys, are below average for the region (PCBS 2021). Children experience frequent disruption to schooling due to recurring crises and conflict and the education system is chronically underfunded resulting in a lack of teachers, facilities and resources (NRC 2021; PCBS 2021; UNICEF, 2022). Palestinian students and educators in the West Bank face daily mobility restrictions due to a complex network of checkpoints, and the separation wall which leads to delays, detention, and harassment by Israeli soldiers and settlers. These barriers limit their freedom of movement and access to education (Friends of Birzeit University (FOBZU),), 2017).

Universities and UNRWA technical training centers are responsible for delivering teacher education programs through education diplomas, bachelor's degrees, and master's degrees. Practicum in Palestinian universities aims to engage preservice teachers in practical teaching experience in schools so they can practice teaching and learning tasks as real schoolteachers (Ayyad, 2013). However, the credit hours and the number of practicum courses differ from one university to another. Most universities offer two practicum courses within their diploma programs; in bachelor programs they provide from four to five practicum courses. The first courses focus on observation and micro-teaching, while advanced practical courses include the application of full lessons. Silmody (2024) recommended the necessity of reconsidering pre-service teacher preparation programs in faculties of education at Palestinian universities in light of the professional standards for Palestinian teachers as identified by the Commission for Developing the Teaching Profession (2012) and emphasizing the integration between the theoretical and practical courses considering the continuity and sequences of the concepts.

The Ministry of Education and Higher Education's (MoEHE) Strategic Plan (2017–2022) emphasized student-centered learning and active learning approaches for developing educational programs, curricula and evaluation systems. Within this framework, PBL is viewed as a promising pedagogy, particularly for preschool and elementary education. Several educational non-governmental organizations (NGOs), such as the Early Childhood Resource Center, Right To Play, and the Qatan Center for Educational Research and Development have implemented projects to promote PBL in the West Bank and Gaza by providing professional development programs for in-service teachers. However, the integration of PBL is often confined to individual initiatives, pilot projects and externally funded programs such as the Learn-by-Play program (UNESCO, 2015). Previous research by our team demonstrated that although teacher educators from various Palestinian universities recognize the value of PBL and were motivated to embed it within their practices, several issues such as insufficient knowledge and skills, cultural norms, and physical barriers limited the opportunities for widespread adoption in classrooms (Khalil et al., 2022; Murtagh et al., 2022).

Additionally, the MoEHE's strategic plan (2017–2022) prioritizes eliminating gender disparities, empowering all women and girls, evaluating school textbooks for gender sensitivity, and ensuring equal access to education at all levels (MoEHE, 2017). There are already higher enrollment and completion rates for females at secondary and tertiary level education in Palestine, compared to their male peers; though this does not

continue through to the labor market where only 19% of women are active (PCBS, 2023; Walsh et al., 2025). The adoption of GRP is in its infancy. The research completed during first phase of this project indicated a significant gap in teacher educators' GRP knowledge and skills, as well as the influence of socio-cultural backgrounds and prevailing cultural norms within their educational contexts on implementation of GRP (Khalil et al., 2024). The broader socio-cultural landscape in Palestine includes patriarchal traditions and religious influences on issues such as underage marriage and gender roles (Hattab & Abualrob, 2023), and the impact of the Israeli occupation on Palestinian living conditions which disempowers women (Gultom & Marulitua, 2024). This further complicates the integration of GRP into educational settings. Although the adoption of both PBL and GRP is still incipient, the research completed during the first phase of this project demonstrated that while teacher educators are motivated to explore these innovative pedagogical approaches, they need to develop their capacity – including knowledge and skills – to adopt gender-responsive and play-based approaches (Khalil et al., 2022; 2024). This research contributes to ongoing educational reform efforts in Palestine aimed at fostering a student-centered, equitable, and contextual grounded learning environment.

Research exploring the influence of teacher educators' teaching practices on preservice teachers' practices is vital given its importance for enhancing the quality of teacher educator programs. While the literature provides evidence of the use of PBL and GRP, both pedagogies are examined independently. The former mainly applied within early childhood and primary education and with limited implementation in secondary education and higher education. Although teacher educators in Palestine, express motivation to apply PBL and GRP within their teaching practices, they encounter significant challenges including insufficient knowledge and a lack of physical and social support (Khalil et al., 2022; 2024).

This study, therefore, aims to understand preservice teachers' practices in embedding PBL and GRP, and is guided by the following research questions:

- How do preservice teachers' capability (psychological and physical skills), opportunity (physical and social contexts) and motivation (reflective and automatic processes) influence their integration of PBL into teaching practices?
- How do preservice teachers' capability, opportunity and motivation shape the ways in which they incorporate GRP in their classroom instruction?

Materials and methods

This study employed a qualitative case study design (Creswell & Creswell, 2017) including semi-structured interviews and observation to understand preservice teachers' practices in incorporating PBL and GRP. We also explored preservice teachers' perceptions of the role of teacher educators who received professional development of PBL and GRP on the enhancement of knowledge, skills, beliefs and motivation for the adoption of PBL and GRP among preservice teachers.

The participants in this study were preservice teachers who were taught by teacher educators who attended the professional development program on PBL and GRP.

The professional development program consisted of a two-day workshop and one-day follow-up session focused on supporting teacher educators to embed PBL and GRP in their courses. The training provided them with practical examples, designing action plans for implementing PBL and GRP in teacher education courses and discussing the challenges of applying both pedagogies in higher education. These teacher educators were encouraged to apply the new pedagogical approaches in their teaching.

A purposive (criterion-based) sampling was employed for the recruitment of pre-service teachers in the third phase of the project, which corresponds to the data and findings of this paper. Participants were intentionally selected based on predefined criteria: First, we chose six teacher educators from the four participating universities who attended the workshops and actively implemented PBL and GRP into their practices among four universities. Second, we asked the selected teacher educators to choose 3–4 preservice teachers per university who would be carrying out their practicum during the 2022/2023 academic year, who had previously engaged in PBL and GRP activities with the selected teacher educators during the preceding semester and who are willing to participate in the study. A total of 17 participants from four universities were selected (4 males, and 13 females).

The research team considered 17 to be a feasible number of participants for both observation and follow-up interviews given practical constraints and methodological considerations of the study including the following factors: (1) the selection of pre-service teachers largely depended on the discretion of the teacher educators who had participated in the professional development program and subsequent interviews with the research team, as well as the willingness and availability of pre-service teachers to participate; (2) the observation took place in the context of ongoing pandemic-related disruption; (3) the ongoing occupation of the West Bank represented significant challenges for travel between regions and work/school and home for both researchers and pre-service teachers alike; (4) two researchers from the team of four were designated to carry out all observation and interviews and (5) the team was concerned with the extent to which preservice teachers across all four universities incorporated the PBL and GRP strategies they had learned from participating teacher educators in their practicum, rather than focusing on the practices of individual preservice teachers. With such practical constraints sample saturation, which could be understood as observation of all pre-service teachers who received classes from participating teacher educators across all four universities, was neither feasible nor necessary for the purposes of data collection.

Their academic specializations were distributed as follows: Physical Education (4), English Language Teaching Methods (2), Special Education (2), and Early Childhood Education (2). The selected preservice educators taught across a range of subjects including math, science, Arabic language, physical education, and English for students in elementary schools. Fourteen participants were in their fourth academic year, while three were in their third academic year. Two preservice teachers had previously engaged in PBL during informal training. Additionally, four preservice teachers were undertaking the practicum for the first time. The gender breakdown reflects the fact that the majority of preservice teachers in teacher education programs in Palestine are females. Two members of the research team (a postdoctoral researcher and research assistant)

visited eight schools and observed all 17 preservice teachers. An interview was conducted with each preservice teacher following the observation of a lesson.

Ethics approval was received from the Research Ethics Committee at Birzeit University (220907) and permission to access the schools was granted by the Ministry of Education and Higher Education. All participants provided informed consent for the interviews and observation.

Research instruments

Observation of lessons was used to explore preservice teachers' practices in implementing PBL and GRP during their practicum in schools. Based on previous studies (Birdwell et al., 2016; Morales et al., 2016; Rimer, 2018), we designed a lesson observation form focusing on the following components: learning activities, encouragement, dialogue, participation opportunities, engagement, feedback, management and class atmosphere (Appendix A). The observations took place within the regions where the participating preservice teachers and the universities were based and included both government primary schools (8) and private primary schools (9). The classes which were observed varied from mixed classes (7), female-only classes (6), and male-only classes (4) across different subjects including Math, Arabic Language, English Language and Physical Education. The research team did not interfere with the identification and selection of the subject of the session, as this was scheduled by the teacher educators and their preservice teachers.

Two researchers conducted in-person observations of the preservice teachers during one lesson of approximately 40 min and took notes directly using the lesson observation form. Each preservice teacher was observed during a 40-minute lesson, which corresponds to the standard instructional period of a complete class within the Palestinian school system. It represents a complete and a coherent unit of teaching practice including the lesson introduction, (interactive) learning activities and closure. The observations were purposefully focused rather than longitudinal, and the researchers acted as passive observers and did not interfere with or involve themselves in the preservice teachers' practices during the lessons. Simultaneously, the lessons were video recorded to generate observation transcripts. After each session, the researchers individually reviewed the video recordings and added further details to the components of the observation form. They then discussed their notes to consolidate the data into a single observation record for each session.

The researchers followed up on the observation through semi-structured interviews with the participating preservice teachers. These interviews were structured around the COM-B model (Michie et al., 2014). The semi-structured interviews were based on a series of open-ended questions designed to understand preservice teachers' knowledge, beliefs, motivations and future intentions, desires, feelings, enablers and challenges for integrating PBL and GRP into their practice. The main questions are presented in Appendix B. Two researchers carried out in-person interviews with the preservice teachers which lasted about 30 min. The interviews were conducted in Arabic and were audio recorded to assist the verbatim transcriptions. The Consolidated Criteria for Reporting Qualitative Research (COREQ) guidelines were followed to report the important information from the interviews (Tong et al., 2007).

Data analysis

The data collected from the interviews and observations was analyzed in four stages using the thematic analysis approach put forward by Braun and Clarke (2006). First, the semi-structured interviews were transcribed verbatim, then both researchers coded the data using English terms and classified the ideas into themes and subthemes individually, and then they met to discuss the common subthemes and resolve inconsistencies in the data. In the second stage, two researchers read each category of observation notes and generated codes. They then drafted the initial themes and then these themes were shared with the wider research team to agree on common findings from the observation. Afterward, the findings from interviews and observation were compared, and the overall findings were determined as a result of merging the findings of interviews and observation. Finally, the resulting themes were classified based on COM-B components of capability, opportunity and motivation. Figure 1 presents an example of how the analysis has been conducted to get the results.

This process was enhanced by collaborative discussions with the research team to reach agreement on the results. Disagreements were resolved through open dialogue, referencing the data, revisiting research objectives, and ultimately reaching a consensus. For example, when some preservice teachers mentioned the school administration as a factor affecting their implementation of PBL, we discussed whether this should be considered a physical or social opportunity. After revisiting the data, we confirmed they were referring to the school administrators rather than policies and thus categorized the issue as social opportunity. The examples from the lesson observations and the quotes from interviews were used to support the results identified by letters and numbers (e.g., X1, X2, …., X17) to ensure participants' anonymity.

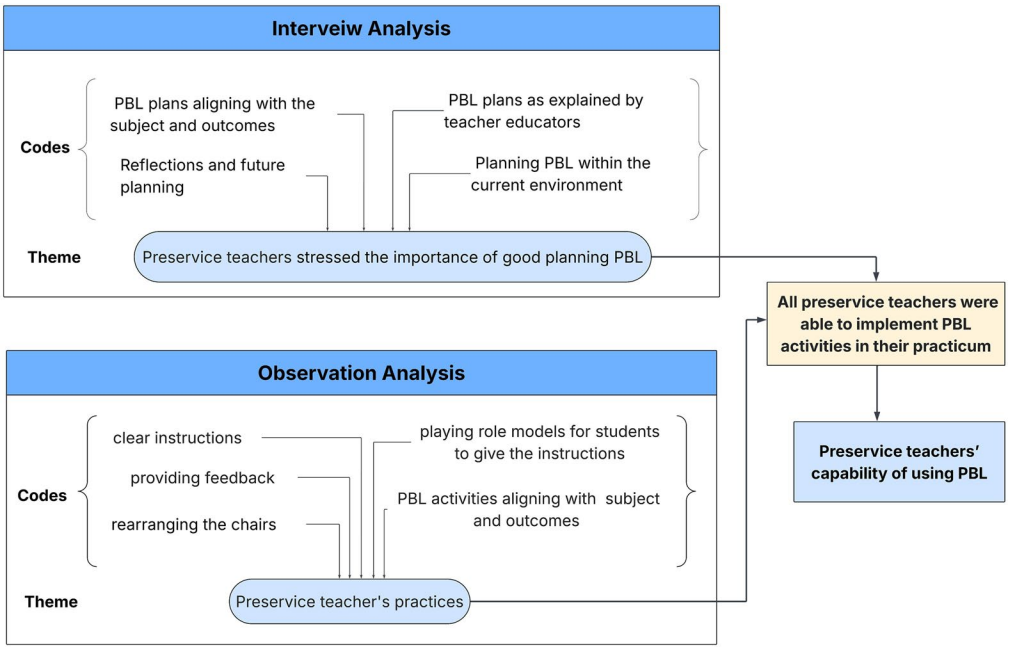


Figure 1. An example of data analysis.

Results

The main findings indicate that preservice teachers were able to use PBL and activities during their practicum, however, they lacked knowledge of GRP despite considering GRP in their practices. Additionally, preservice teachers could identify the physical and social opportunities during their implementation, we found that teacher educators play a significant role in enhancing preservice teachers' practices of PBL in terms of knowledge, skills and motivation. Finally, all preservice teachers believed that PBL is an essential strategy for learning and demonstrated automatic motivation toward using PBL. The results are presented based on the key research questions and are organized according to the main components of the COM-B model.

Understanding preservice teachers' practice in embedding PBL

The following findings address the first research question concerning preservice teachers' practices in incorporating PBL. The results are organized into three subsections based on the COM-B components.

Capability (knowledge and skills)

The findings indicate that preservice teachers described PBL and its diverse benefits. Some preservice teachers understood PBL as a nontraditional strategy, while others described PBL as an enjoyable strategy that helps the students to concentrate, stay active, and expand their knowledge easily. Most of them confirmed that PBL motivates the students to learn and be more active, and it is a cooperative method of learning. One preservice teacher reported that "PBL motivates the students' emotions towards learning by loving school and liking me as a non-traditional teacher" (X4). Preservice teachers specializing in physical education defined PBL as a strategy that releases energy and helps to prepare students for the physical aspects of learning new motor skills needed in physical education, in addition to enhancing teamwork.

Additionally, all preservice teachers were able to implement PBL activities. We observed that most preservice teachers began their activities by providing clear instructions to the students, and three preservice teachers acted as role models to demonstrate the steps of the activity. All preservice teachers were able to choose appropriate PBL activities for the subject and for achieving their learning goals. In the interviews, they explained that the content and outcomes of the lesson are the principal factors which should be considered when deciding which PBL activity should be applied as this was something they had discussed with their teacher educators. One preservice teacher commented: "I believe that the subject, contents of the lesson, and which goal I need to reach from the lesson determine the kind of activity I apply with students." (X13). This is also applied to special education classes, where preservice teachers used drama to present the challenges of disabilities within social contexts, while preservice teachers teaching math used guessing games to demonstrate odd and even numbers. Additionally, preservice teachers provided their students with feedback during and after implementing PBL activities through immediate feedback or peer feedback.

Although the school learning environment had limitations such as a lack of space, a large number of students, and a shortage of play tools, all preservice teachers were

able to apply PBL activities within the physical context. We observed that eight preservice teachers rearranged the chairs and the tables in the classroom. Most preservice teachers confirmed during their interviews that they should apply PBL activities according to the possibilities within the physical environment (space, number of students, availability of tools), as they learned from their teacher educators, and that they need to have good plans for implementing PBL activities within the physical and social context of each classroom and school. One preservice teacher commented that they "learned that teachers could apply any PBL activity under any conditions if they want to do that, they need good planning within the current situation." (X6). Another preservice teacher proposed a practical solution to spatial limitations, stating, "if the classroom space is limited, I will conduct PBL activities in the schoolyard or reserve a larger classroom for these activities." (X10). Several preservice teachers also reflected on their PBL teaching experiences and suggested alternative activities/procedures for future implementation. For instance, one preservice teacher shared that "I conducted PBL activities that required students to use their hands. I didn't know at the time that one student couldn't use his hands. Next time, I will plan an activity that allows him to participate with his classmates." (X2).

Motivation (reflective and automatic)

Preservice teachers are highly motivated to use PBL in their teaching practices, we found that: (1) Reflective Motivation: All preservice teachers believe that PBL is an important strategy to use in their practices (2) Automatic Motivation: All preservice teachers were happy and excited when applying PBL.

Reflective motivation. All preservice teachers stressed the importance of using PBL in their future teaching practice at schools. Preservice teachers mentioned that they were excited when their teacher educators used PBL during their learning classes at the university, and when they applied PBL strategies in the practicum at schools. "I believe that when the students are excited this reflects on my emotion to be motivated for teaching." (X14). Most preservice teachers said that PBL activities suit all students' ages, including adults, while two mentioned that it suits only primary school children.

Automatic motivation. All preservice teachers were excited and happy when applying PBL in their teaching. The students' emotions that we observed when they were conducting PBL in the classroom included happiness and excitement. Additionally, preservice teachers expressed their happiness during the interviews. One preservice teacher commented "It is a wonderful strategy, when the students are sleepy, I use this activity to make them active and change the class environment to become interactive. This makes me happy and excited while teaching PBL." (X6). Although four physical education preservice teachers expressed that they were happy conducting PBL activities, they also mentioned that they found it stressful as it was their first time using PBL.

Preservice teachers commented that positive reactions from students encouraged them to adopt PBL in their future teaching practices. The researchers noticed a positive atmosphere in the classrooms when preservice teachers conducted PBL including excitement, happiness, laughing, concentration, high interaction, sharing, and

cooperation. This is shown from the observation where, in the Math classroom, for example, the preservice teacher (X10) found the students were overcome with excitement, a response that the preservice teacher had not anticipated and would have to prepare for this level of enthusiasm in future classes.

Opportunity (physical and social)

Regarding the opportunity for implementing PBL, preservice teachers identified various enablers and challenges of applying PBL in teaching at school. We classified these findings based on the COM-B model as physical and social opportunity as follows:

Physical opportunity. Preservice teachers mentioned that the physical factors that would help them in implementing PBL in their teaching at school include: A smaller number of students in the classroom, good planning for PBL, classroom management and availability of PBL tools at school. Most preservice teachers also noted similar areas as current challenges to implementing PBL during practicum, including large class sizes, shortage of resources, and lack of space in the classroom. A preservice teacher (X10) commented "Sometimes, PBL activities are limited to a certain number of students, or class size is large, and the space is small. Tools may also be unavailable, the school offers simple tools, so I have to find a balance between all available resources."

Social opportunity. The main social factor that affects preservice teachers' practices is the influence of the teacher educators. This study found that teacher educators influence preservice teachers' practices in three ways: (1) Preservice teachers' knowledge and skills of using PBL, (2) planning of PBL activities, and (3) motivation to apply PBL. Our findings in this section are as follows:

The knowledge and skills of preservice teachers regarding PBL were enhanced by their teacher educators' practices of PBL during the previous semester. All preservice teachers mentioned that their teacher educators had discussed and practiced PBL with them during the previous semester. They learned from their teacher educators how to choose the appropriate PBL activities that align with the learning outcomes and fit the subject. A preservice teacher (X2) commented that "I discussed the PBL lesson plan with my teacher educator and revisited my notes last semester, when we implemented PBL together." Additionally, most preservice teachers mentioned that they have learned how to adapt PBL activities within the classroom context, including the class space, the number of students, the availability of the tools, time, and management. Some preservice teachers mentioned that they regarded their teacher educators as role models in the application of PBL.

Preservice teachers confirmed the significant role teacher educators have when it comes to planning their PBL activities. For example, one preservice teacher (X1) mentioned that his teacher educator advised them to use colors and shapes and take into consideration special needs in the classroom. Furthermore, their teacher educators stressed the importance of good planning for PBL activities within the school context. This is clear when preservice teachers designed their PBL activities within their physical environment and responded to the challenges presented at different schools. While teacher educators were the principal reference for preservice teachers in planning PBL

activities, preservice teachers also used other resources such as discussing with families and classmates, searching online, asking the original schoolteacher, and reviewing university curriculum. One preservice teacher (X4) commented that "I got the idea of using balloons from my teacher educator last semester, and I discussed implementing this PBL activity beforehand. I also search online to find practical examples implemented by schoolteachers to learn from their experiences."

Preservice teachers' motivation for using PBL was impacted by their teacher educators. All preservice teachers were excited and highly motivated to use PBL in their future practices because they were excited when their teacher educators used PBL during their learning classes at the university, and their teacher educators highly encouraged them to use PBL in schools. A preservice teacher (X12) explained "I was happy when my teacher educator implemented traditional games such as hopscotch and dice. This made me think if we enjoyed them, how much more would the children at school?". This is an example in itself of how the use of PBL supports active learning throughout all levels of education, including university.

In addition, preservice teachers identified school students' profiles as another social factor which affects their practices in embedding PBL at school. These include previous knowledge about the students' background at school, choosing a PBL activity that suits students of all levels including students with disabilities, individual differences and language differences, and students' motivation toward PBL, one preservice teacher commented that "When I see that the students inside the classroom are motivated, this encourages me to conduct more PBL activities."(X8). In addition, the results showed that seven preservice teachers agreed that student knowledge, beliefs and skills about PBL affect their implementation of PBL activities, four of them identified a challenge in the fact that PBL is a new experience for the students at that school "I faced some challenges when I applied PBL for the first time for students because they weren't used to it, so I was confused about how to deal with these cases."(X14)

The challenges included a lack of management skills in the classroom, cooperation from the school administration, and interference from the original teacher. Regarding the latter, one preservice teacher commented on how "the original teacher refused to let me conduct PBL activities because they thought it was a waste of time." (X6). Regarding school administration, some preservice teachers reported that the school administration supported their efforts of implementing PBL. One preservice teacher stated that "the school administration encouraged me in applying PBL in teaching English for elementary schools" (X16). However, two preservice teachers identified the school administration as a barrier to implementing PBL. For instance, a preservice teacher remarked, "I asked for some tools to use in PBL, and they refused to support me." (X17).

Understanding preservice teachers' practices in incorporating GRP

The results in this section respond to the second research question regarding preservice teachers' practices in embedding GRP which are classified into three categories:

Capability (knowledge and skills)

The interview results showed that preservice teachers gained insufficient knowledge of GRP as a strategy which can be adopted alongside PBL. This lack of knowledge

impacted preservice teachers' capability and motivations for incorporating GRP into their teaching practices relative to PBL. When asked about their understanding of GRP some expressed an awareness of the importance of creating gender equity in the classroom, but others did not have a clear idea of what GRP might involve or did not believe it was relevant to their teaching. For instance, two preservice teachers described GRP as a way of integrating females and males during the teaching and learning process and ensuring their equal participation in implementing the activities with due consideration for other special cases, such as students with disabilities. Five preservice teachers mentioned that GRP relates to the consideration of individual differences when choosing PBL activities, and another five participants stressed that they should consider the differences between genders when using PBL. One preservice teacher commented "I think GRP involves considering the individual differences among students, so I selected a PBL activity that accommodates all learners" (X13). However, five preservice teachers out of 17 said that they did not know what was meant by GRP strategies.

Despite the lack of understanding or confusions expressed about the nature of GRP in their interviews, we observed that preservice did use some GRP approaches in their practices. For instance, all the preservice teachers we observed who applied PBL in mixed gender classes made sure female and male students were treated equally during the activities. It was also clear from the observation that they provided equal opportunities for the participation of both genders in PBL activities and encouraged all of them to be active in the classroom. We noticed, for example, that one preservice teacher encouraged the only female student in the group to participate in the activity with her male classmates.

Opportunity

Preservice teachers did not specifically raise any issues in relation to physical or social opportunities to implement GRP during the interviews. We suspect this may be because of their limited experience of GRP, which may have left them insufficiently prepared to critically reflect on the concept. The pre-service teachers demonstrated basic gender-responsive practices, consistently organizing students into mixed-gender groups with ease. Throughout the implementation of class activities, no instances of single-gender grouping were observed.

Motivation

During the interviews, most preservice teachers emphasized the importance of ensuring equal participation for all students when planning and implementing PBL. However, they did not explicitly refer to GRP as they don't have sufficient knowledge about it. One preservice teacher reported that "it is important to implement these pedagogies especially in creating the groups" (X15).

Discussion

The results of this study suggest that the preservice teachers who received classes from the teacher educators have adequate knowledge and skills to implement PBL in their

practicum. It also demonstrates how they were able to choose appropriate PBL activities in line with the subject and learning outcomes, as well as adapting PBL activities to the physical contexts of the classroom. The results demonstrate a general lack of comprehension of GRP, even though preservice teachers did demonstrate some adoption of GRP practices during practicum. Additionally, the study also identifies physical and social opportunities that preservice teachers encountered while conducting PBL in their practicum. The results reveal the important role of teacher educators in influencing preservice teachers' practices of PBL in terms of knowledge, skills and motivation, and school students as social factors that affect their practices. Finally, the results indicate that preservice teachers recognized the great importance of using PBL in their practicum and future teaching practices in school.

Our interpretation of the findings suggests there is a relationship between teacher educators and preservice teachers in terms of knowledge, skills and motivation for using PBL and GRP. Preservice teachers explained that their fruitful experience with teacher educators who used PBL in their classes encouraged them to adopt this approach in the future. This confirms the result of the positive relationship between teacher educators' practices and preservice teachers' intrinsic motivation (Chan et al., 2023). Therefore, preservice teachers should be exposed to new approaches at university so they can create meaningful learning experiences with their students in school. Furthermore, this study provides evidence to support the argument that teacher educators should implement new strategies with preservice teachers in their classes.

The results showed that the participants were motivated and had the necessary knowledge and skills to implement PBL in their practicum and expressed their excitement when their teacher educators applied PBL with them. Self-determination theory is useful for understanding how, when individuals are satisfied and interested, they are open to implementing new actions and developing their skills (Ryan & Deci, 2000). This is aligned with the study of Jensen and Rørbæk (2022) which demonstrated that PBL is an invaluable strategy for preservice teachers' enjoyment while studying in teacher education programs. Additionally, this result indicated that preservice teachers believe that PBL is an essential strategy for students of all levels, not only for school students but also for university students. This confirms the results of previous studies that stressed the importance of using PBL in a range of subjects in higher education, for example, using PBL with undergraduate and master students in engineering education in the U.S. (Kothmann, 2019), and improving undergraduate business students' motivation and absorption in an Australian university (Khan & Pearce, 2015). Furthermore, the results indicated that positive students 'reactions motivated preservice teachers to adopt PBL within their future teaching practices, as preservice teachers associated the student engagement and enjoyment with the benefits of PBL. Other studies have shown that preservice teachers appreciate the fulfillment, enjoyment and joy from witnessing their students have fun when participating in PBL activities and consider it as a meaningful and effective teaching experience for both students and teachers (Saddaramil et al., 2025). Moreover, the results indicated that the participants were able to implement various PBL activities in a range of subjects considering the social and physical contexts. More research is needed to deepen our understanding of how and why preservice teachers choose these PBL activities. Without a background in PBL and adult learning, PBL in higher education will be complicated (Whitton, 2018).

The results offer new evidence for the strong connection between preservice teachers and teacher educators. Previous studies investigated this connection in terms of teacher educators' instructional behaviors and preservice teachers' intrinsic learning motivation (Chan et al., 2023), teacher educators' values and their preservice teachers in using simulated teaching experiences (Mikeska et al., 2023), and teacher educators and preservice teachers in modeling a learning-centered lesson design approach (Gravett & van der Merwe, 2023). This study contributes further to the evidence that demonstrates the relationship between teacher educators' and preservice teachers' knowledge, skills and motivation regarding using PBL. This implies a need to enhance teacher educators' practices through professional development programs focused on innovative teaching methods which will enhance the quality of teacher education programs. We recommend that these results be adopted by the faculties of education in teacher education programs and reflected as actions in the updated version of the teacher education strategy and MoEHE strategic plan for developing preservice teacher education in Palestine.

One point to note is that preservice teachers' comprehension, motivation and adoption of GRP continue to be limited. Previous studies have suggested that a contributing factor could be the fact that the majority of preservice teachers in the faculties of education are females, resulting in fewer opportunities for teacher educators to explicitly demonstrate all aspects of GRP with preservice teachers in their modules. Due to the lack of practical experiences with GRP during their teaching education programs, preservice teachers struggle to explain and apply its principles effectively. This was previously noted in the study by Ananga (2021) which indicated that large class sizes and the disproportionate numbers of female than male pre-service teachers in the same class limited the opportunity to practice GRP in class. This pattern was also evident in our study, which included a similarly uneven gender distribution with 13 female and 4 male pre-service teachers. This is aligned with the general composition of teacher education programs in Palestine - and many other countries - where female preservice teachers typically outnumber their male counterparts. In this context it may have been challenging for teacher educators to model - and for preservice teachers to experience - a broad range of gender-responsive practices during taught elements of the teacher education program. Another reason for the limited use of GRP could be related to the need to improve teacher educators' knowledge of GRP to be able to embed this pedagogy in their courses and to generate open discussions about GRP with their pre-service teachers. In a study conducted by Alemayehu (2020) they found that university teachers did not apply GRP in their lesson plans due to the lack of specific training on GRP. In Palestine pre-service teachers often lack opportunity to discuss cultural norms and biases related to gender which hinders the development of their knowledge of gender equality and GRP. Additionally, gender issues are often treated as secondary topics rather than essential components in teacher education curricula. This limited critical reflection on preservice teachers' cultural norms may result in a fragmented understanding and inconsistent implementation of GRP. More work is needed to improve preservice teachers' ability to address gender equality in the educational context as recommended by Şanlı and Karakuş (2023), and further studies focusing specifically on GRP should be carried out to develop a better understanding of GRP among preservice teachers.

A previously published paper (Walsh et al., 2025) discusses in detail how PBL, as a more active and easily identifiable and accessible pedagogical approach, seems to have overshadowed GRP in this project. This is symptomatic of the cultural sensitivities mentioned here and that some of the explicit gender content is lost when used with PBL. The authors (Walsh et al., 2025) suggest that PBL and GRP can be mutually reinforcing approaches, but that it is necessary to ensure teacher educators have training in GRP specifically, so the gender content does not get 'lost in translation.' As the review of the literature in this paper makes clear, the research and guidance around GRP is heavily influenced by international development practice which overwhelmingly focuses on removing barriers to accessing education in the Global South, rather than addressing questions of equity within the classroom, learning environment, learning materials or teaching approaches. The authors believe that an expanded GRP research agenda should consider factors other than barriers to access and into a diversity of strategies for creating gender equitable classrooms including modifications to curricula and materials as well as teaching practice (Baily & Holmarsdottir, 2015; Banerjee, 2024), how best to support pre-service teachers in implementing these strategies once they enter the classroom (Knowles & Hadley, 2025), the potential for feminist pedagogies to be incorporated into teacher education and the wider curriculum and how to incorporate expansive understandings of gender beyond male-female binaries (García & Slesaransky-Poe, 2010).

The research team identified a few limitations with this study. First, this study does not cover all teacher educators nor preservice teachers across all four participating universities. The study was limited to preservice teachers who were taught by teacher educators who received professional development on PBL and GRP during the academic year 2021/2022 and were enrolled in their practicum for the second semester of the academic year 2022/2023. Second, the implementation of PBL and GRP in the practicum for the full cohort of students who took classes with the teacher educators who participated in the professional development program cannot be generalized, as the preservice teachers in this study were chosen based on their availability and willingness to participate. Third, the results might be different across different cultural and educational settings because GRP is context-specific. In Palestine, understanding the interplay between gender, religion, tradition, and socio- political dynamic is complex, and GRP should be framed in a way that aligns with religious and cultural values. Despite these limitations, the study offers valuable insights for the future design of teacher education programs in Palestine and elsewhere. In particular, we demonstrate that PBL can be successfully embedded in a range of subject areas and can enhance the student experience of those studying at universities and in schools.

Conclusion

This study makes a significant contribution to understanding the strategies to support preservice teachers in implementing PBL and GRP in their classrooms. Through the qualitative analysis of observations and interviews we conducted with 17 preservice teachers across four universities in Palestine, we found that preservice teachers demonstrated notable capability and motivation in implementing PBL, supported by the opportunities afforded through their classroom environments and the modeling provided

by teacher educators. In contrast, their limited understanding of GRP meant that both their capability and motivation to use gender-responsive practices were constrained, as they had fewer instances to engage with or apply GRP purposefully, despite some equitable practices emerging unintentionally.

The preservice teachers demonstrated adequate skills to embed PBL activities in their practices in their classes across a range of subjects. We also observed that pre-service teachers demonstrated appropriate planning including designing PBL activities in line with the subject and learning outcomes considering the physical and social context of the classroom.

Additionally, the results have shown the significance of teacher educators' role in preservice teachers' knowledge, skills and motivation toward PBL and GRP. When preservice teachers experience new strategies such as PBL during the learning process in teacher education programs and recognize the usefulness of this strategy in teaching and learning, they are motivated to apply this strategy with their students in school. This result suggests that teacher educators should apply evidence-based innovative strategies with preservice teachers. Similarly, when preservice teachers did not receive explicit instruction regarding GRP, they lacked knowledge of GRP. Therefore, more work will be needed to enhance preservice teachers' GRP practices, and further research is required to better understand the development of GRP practices.

The results contribute to an expanded understanding of how to improve the teaching quality of preservice teachers, particularly for PBL. Overall, this study provides further evidence for the value of providing professional development opportunities for teacher educators, given the multiplier effect on preservice teachers' practices. Further research is needed to ascertain how best to mainstream gender issues in teacher education programs.

Our findings have practical implications for teacher educators, university administrators, school principals and the Ministry of Education. We acknowledge that the following recommendations may not be feasible for all stakeholders in all contexts, particularly in settings with limited educational funding and resources, or in countries affected by occupation or conflict. Given our finding that teacher educators are central to shaping preservice teachers' knowledge and application of novel pedagogies, and that modeling greatly influences future practices, we recommend that PBL and GRP are explicitly integrated into modules and university classes. This should include examples of PBL and GRP practices in lesson planning, classroom interactions and assessment. Structured practice opportunities could also be provided for pre-service teachers through micro-teaching and/or video reflection. This could be expanded further through co-designed PBL tasks and GRP scenarios with mentor/cooperating teachers that reflect local context.

It was noted in earlier phases on this project that university management creates institutional conditions that support or constrain innovation in teacher education. In order to ensure that teacher educators are well positioned to model effective practice, we recommend that professional development for teacher educators is mainstreamed to provide ongoing training on PBL and GRP, rather than as once-off workshops. Relevant teaching materials and resources should also be allocated. We note the particular challenges encountered with respect to adopting GPR; therefore, we recommend developing support on integrating GRP into planning and classroom management.

This includes practical examples from diverse contexts, including Palestine, and addresses the cultural norms and biases to support effective GRP implementation. Where in-house expertise does not exist, gender-equity specialists and partner NGOs could provide appropriate training.

Schools shape the practicum environment where pre-service teachers attempt to apply PBL and GRP in "real world" settings. To ensure that preservice teachers receive constructive feedback from mentor/cooperating teachers in relation to the implementation of PBL and GRP, it may be necessary to host discussion sessions between teacher educators, principals and teachers to align expectations. This could include capacity building if these pedagogies are not already embedded in school practice.

We note that strategic policy can mainstream practices that individual universities, programs and teacher educators may find challenging to sustain. The Ministry of Education may play an important role here, by including PBL and GRP competencies in the national standards for initial teacher education. National training programs that scale professional development for teacher educators should also be supported. Furthermore, an open repository containing PBL and GPR resources might be established, providing culturally relevant models for schools. The Ministry is well positioned to strengthen school-university partnerships by offering support for formal collaborations. This would promote a shared understanding of PBL and GRP, and help ensure that these priorities are reflected in teacher education programs and school practices. Lastly, the Ministry may support longitudinal studies examining how PBL and GRP develop over time, including the impact of teacher educators' professional development on the experience of pre-services teachers and ultimately classroom practice. As the Palestinian educational system has been operating under continuing wars and emergencies, integrating PBL and GPR within the Palestinian teacher training offered by the faculties of education and Ministry of Education may support children wellbeing by reducing stress, rebuilding a sense of safety and normalcy and encouraging inclusive participation.

Collectively these practical implications aim to strengthen preservice teachers' education in PBL and GRP by improving modeling, embedding these concepts in curricula, fostering supportive school environments and promoting coherent national policy. These actions could play an important role in ensuring that future teachers embark on their career equipped to deliver meaningful, equitable and innovative learning experience.

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Appendix A

Observation sheet

Preservice teacher observation form: PBL- GR observation sheet

Observer/s: _____

Preservice teacher observed _____Gender of S.T.: _____

University: _____Her/his Teacher Educator: _____

Date: _____Time: _____ Course: _____

Name of school (practicum):_____

Active learning components
All the components of observation are addressing PBL and GR practices of the preservice teacher and shall be included while the observation.

- **Learning activities**
- What kind of activities' tools did the preservice teacher use with students?
- How were the activities applicable for both genders?
- How did the preservice teacher conduct these activities? considering males, females?

- **Student- Teacher's instructions**
- How did the preservice teacher give directions to class activities? How is it for males vs females?
- How did the students follow the preservice teachers' instructions inside the class? (note by gender)
- **Encouragement**
- How did the preservice teacher encourage and ensure all student participation in the activities?
- **Dialogue & Language**
- How was the dialogue between preservice teacher and students during class activities? Any concentration on males, females? Equal participation?
- What kind of language (examples/quotations) did the teacher use with students?
- **Participation opportunity**
- How did the preservice teacher give opportunity for both genders to interact with her/him and with each other?
- **Engagement**
- How did the preservice teacher engage the students with the activities?
- How did the preservice teacher give equal participation inside the class?
- How did the students engage with other students/teachers?
- **Feedback**
- How did the preservice teacher provide feedback on the activity to the students?
- **Management**
- How did the preservice teacher distribute group work and activities inside the class? Considering females and males.
- How did the preservice teacher manage the class space in his activity with both genders?
- **Exceptional observations**: Any exceptional observation of the preservice teacher's practice of class activities.
- **Class Atmosphere**: The overall atmosphere of the classroom when the preservice teacher did practice class activities.

Appendix B

The main interview questions

1. How would you describe PBL and GRP in your own words?
2. Describe your experience in Practicum? If so, what specific training have you received?
3. Have you had the opportunity to apply for PBL and GRP during your practicum? If yes, describe your experience.
4. What was your role in implementing PBL and GRP in the classroom? Describe your experience.
5. What facilities were available when you applied for PBL and GRP within the classroom? How did you contribute to facilitating the learning process?
6. What challenges did you encounter during the implementation? How did you address or overcome these challenges?

7. How did you plan your lesson incorporating PBL and GRP? What resources supported your planning?
8. How was the classroom atmosphere when implementing PBL and GRP?
9. How did the students engage during the PBL and GRP lessons?
10. How did you feel during the session where PBL and GRP were implemented?
11. In your opinion, what is the significance of using PBL and GRP in schools in the future? What kind of support would help you implement these approaches more effectively? How would you plan to support the integration of PBL and GRP into teaching?